

CAMROSE, AB, Canada

WMO: 712540

Lat: 53.0428N

Lon: 112.8217W

Elev: 739

StdP: 92.76

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.1	(c) -28.8	(d) -35.4	(e) 0.1	(f) -32.0	(g) -31.9	(h) 0.2	(i) -28.6	(j) 10.9	(k) -4.5	(l) 9.8	(m) -7.1	(n) 1.8	(o) 270	(p) 0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	28.2	18.1	26.4	17.3	24.8	16.5	20.0	25.9	18.7	24.3	17.6	22.9	4.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	14.2	22.6	16.7	13.0	21.1	15.6	12.1	20.0	60.5	25.8	56.2	24.3	52.7	23.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.2	(b) 8.8	(c) 7.7	(e) -37.1	(f) 31.6	(g) 4.6	(h) 1.8	(i) -40.4	(j) 32.9	(k) -43.0	(l) 34.0	(m) -45.6	(n) 35.0	(o) -48.9	(p) 36.4		
(4) 10.2	(5) 8.8	(6) 7.7	(7) -37.0	(8) 22.3	(9) 4.4	(10) 1.3	(11) -40.2	(12) 23.2	(13) -42.8	(14) 24.0	(15) -45.2	(16) 24.8	(17) -48.4	(18) 25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-12.2	-10.6	-5.2	3.6	10.2	14.2	16.7	15.5	10.6	3.7	-4.9	-11.0		
	DBStd	12.08	8.96	7.81	7.95	5.20	4.45	3.12	2.97	3.24	4.33	4.74	6.90	8.01		
	HDD10.0	3337	688	578	470	197	53	4	1	2	44	201	448	651		
	HDD18.3	5769	947	811	728	441	253	128	68	98	233	455	698	909		
	CDD10.0	640	0	0	0	6	58	131	208	171	62	4	0	0		
	CDD18.3	32	0	0	0	0	1	4	17	9	1	0	0	0		
	CDH23.3	696	0	0	0	1	54	91	266	212	71	2	0	0		
	CDH26.7	141	0	0	0	0	8	13	61	44	14	0	0	0		
	WSAvg	3.5	3.1	3.1	3.5	4.1	4.2	3.8	3.3	3.1	3.4	3.6	3.2	3.1		
	PrecAvg	499	29	21	29	21	48	86	85	70	45	17	19	25		
	PrecMax	776	68	47	86	69	145	217	171	170	142	53	45	60		
(15) Precipitation	PrecMin	349	3	1	2	0	10	11	25	16	3	0	1	8		
	PrecStd	97	19	13	23	14	30	52	40	45	32	14	12	13		
	DB	6.5	7.1	12.7	22.1	27.8	28.4	31.0	30.4	28.8	21.7	12.8	7.4			
	MCWB	3.0	3.2	6.7	11.0	14.4	17.4	20.0	18.7	16.8	12.7	6.9	3.5			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	4.4	4.5	9.0	18.9	24.7	25.9	28.3	27.7	25.3	17.3	8.4	4.7			
	2% MCWB	1.9	1.6	4.3	9.7	13.2	15.9	19.5	18.2	15.1	10.4	4.5	1.9			
	5% DB	2.4	2.4	6.6	15.6	22.2	23.8	26.3	25.7	22.5	14.2	5.8	2.4			
	5% MCWB	0.6	0.2	3.1	7.7	12.2	15.2	18.5	17.6	14.1	8.3	2.8	0.2			
	10% DB	0.2	0.4	4.5	12.8	19.7	21.7	24.3	23.7	19.5	11.6	3.6	0.3			
	10% MCWB	-1.3	-1.1	1.9	6.4	10.9	14.4	17.6	17.0	12.8	6.8	1.4	-1.3			
	0.4% WB	3.4	3.5	7.1	11.7	15.6	19.3	22.3	21.2	17.5	13.1	7.3	4.0			
	0.4% MCDB	5.8	6.6	11.5	21.0	23.2	25.6	27.6	26.2	25.8	20.6	12.1	6.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	2.1	1.8	4.8	10.1	14.1	17.5	20.5	19.6	15.8	10.7	4.8	2.0			
	2% MCDB	4.2	4.0	8.2	17.7	22.7	23.1	26.3	25.4	23.6	16.3	7.9	4.4			
	5% WB	0.7	0.4	3.3	8.4	12.9	16.4	19.2	18.4	14.6	8.9	3.1	0.4			
	5% MCDB	2.4	2.2	6.2	14.7	20.8	21.7	24.9	24.1	21.5	13.3	5.4	2.3			
	10% WB	-1.3	-1.2	2.0	6.9	11.7	15.2	18.2	17.3	13.2	7.3	1.6	-1.3			
	10% MCDB	0.1	0.4	4.3	12.0	18.6	20.2	23.3	22.9	18.8	11.0	3.3	0.3			
	MDBR	10.6	11.8	10.8	11.9	13.2	11.7	12.4	13.2	13.3	11.9	9.7	10.1			
	5% DB	11.9	13.9	13.0	17.2	17.4	15.5	15.8	17.4	19.3	16.7	12.1	11.3			
(36) Mean Daily Temperature Range	MCWBR	9.8	11.3	8.9	9.2	7.9	7.2	8.0	9.0	9.8	10.0	8.5	8.8			
	MCDBR	11.8	13.4	12.4	16.1	15.8	13.1	14.3	15.6	17.6	15.8	11.6	11.2			
	MCWBR	9.9	11.0	8.8	9.0	7.5	6.9	7.8	8.6	9.3	9.8	8.3	9.1			
	MCDBR	11.8	13.4	12.4	16.1	15.8	13.1	14.3	15.6	17.6	15.8	11.6	11.2			
(40) Clear-Sky Solar Irradiance	taub	0.261	0.260	0.302	0.328	0.345	0.356	0.355	0.365	0.310	0.280	0.253	0.246			
	taud	2.298	2.385	2.362	2.372	2.377	2.369	2.405	2.344	2.478	2.537	2.452	2.289			
	Ebn at Noon	734	870	891	907	904	894	889	856	875	840	759	682			
	Edh at Noon	59	74	95	107	112	115	109	109	84	64	50	49			
(44) All-Sky Solar Radiation	RadAvg	1.03	2.08	3.56	4.70	5.74	6.05	6.26	5.14	3.56	2.08	1.03	0.75			
	RadStd	0.11	0.17	0.27	0.45	0.39	0.39	0.31	0.35	0.37	0.24	0.11	0.07			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (87 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page